Rho estimation options — plain-language guide

For: Charles (edit freely; discuss with Alex)
Last synced: 2026-08-17
Context: Paper Directions 21 — fit homophily knob ρ (rho) on **empirical NCAA rosters**, separate from draft maximum likelihood estimation (MLE)
Companions: - `../Alex_stuff/MLE_basics.md` — Bernoulli draft MLE (λ, t); **no** ρ on fixed rosters - `../Alex_stuff/PD20_K_draws_and_rho_explainer.md` — why ρ and draft likelihood are different layers - `../../transcripts/PD21_notes.md` — Alex lock (Aug 14) - `../re_entry/HEROs_and_PASSes/Alex_LG_three_step_briefing.md` — Grandchild assignment (LG) formulas
Locked project choices (Aug 14): - **Assignment engine:** Grandchild assignment (LG) only — not Pass B soft assign - **Data:** empirical NCAA, empirical roster caps, 2011–2021 - **Outputs:** one ρ^* per season + one **longitudinal** ρ^* pooled across seasons - **Seeds:** 50 simulation runs per (ρ, season) to start (increase later; high-performance computing (HPC) when stable) - **Success (stronger):** match sorting index (H_{sort}), global within-team sum of squares (global_wss), and team congestion (L_C) distribution vs empirical; report vs legacy $\rho = 0.5$

What all four options are trying to do

You have **real NCAA men's basketball data** for 2011–2021. For each season you know:

Input	Meaning
$\hat{A}_i$	Player ability (performance per minute, z-scored within season — hero panel)
Roster caps	How many seats each real team has (stub capacity multiset)
Team labels	**Who ended up on which team** (observed assignment)

You also have a **simulation rule** for building synthetic rosters: **Grandchild assignment (LG)**. It seats players one at a time. The homophily knob ρ controls how strongly a player prefers teams whose **current** average ability is close to their own.

The fitting problem: find the value of ρ that makes Grandchild assignment (LG) produce rosters that **look like** the real ones — using **team assignment data only**, not draft outcomes.

Important: none of the four options below uses "was this player drafted?" That is a **separate** step (Bernoulli softmax draft MLE for λ, t).

Shared vocabulary

Term	Plain meaning
ρ **(rho)**	Homophily knob in Grandchild assignment (LG). Higher ρ → players sit with more similar teammates. $\rho = 0$ → nearly random seating among open teams.
Sorting index (H_{sort}**)**	After rosters are set: how much do team labels explain ability spread? Near 0 = teams look random; higher = more ability-sorted. Formula: $1 - \frac{\text{within-team sum of squares}}{\text{total sum of squares}}$.
Global within-team sum of squares (global_wss)	Raw measure of spread around team means. Same information as H_{sort}, but not scaled to 0–1.
Team congestion (L_C**)**	How crowded a team is with "viable" peers (smooth viability formula). Used in scoring later; also a diagnostic for roster structure.
Stub capacity (R_j**)**	Open seats on team j during seating. Grandchild weights multiply homophily by remaining seats.
Random seed	Grandchild assignment (LG) shuffles player order and randomizes team draws. Same ρ, different seed → slightly different rosters.

Panel policy vs roster caps (don't conflate them)

Two design choices look related but solve **different problems**. Discuss them separately with Alex.

Choice	What it controls	Tied to minutes filter?
Minutes / ability policy	Who is on the panel and what $\hat{A}_i$ they carry	**Yes**
Empirical roster caps (R_j**)**	Real NCAA team sizes (heterogeneous stub capacities)	**No**

Empirical caps are not a workaround for filtering. Teams have different roster sizes whether or not you drop bench players. Grandchild assignment (LG) still needs a capacity multiset that sums to N. Moving to fixed roster size C would simplify the engine but **throws away** real NCAA roster-size heterogeneity — a separate modeling choice Alex already locked for PD21.

Default panel (Aug 14 baseline)

- **Filter:** drop player-seasons with **under 20 minutes** (`min_minutes = 20`).
- **Ability:** PPM z-scored within season on the **filtered** panel.

- **Caps:** recomputed from that filtered panel (smaller N, fewer players per team).

Issue we saw: on this panel, **6/11 seasons** had per-season $\rho^* = 0$ and longitudinal $\rho^* \approx 0.07$ — very low assortativity. Simulated sorting at $\rho = 0$ was already close to (or above) empirical on the **trimmed** league.

Alternative panel (Aug 17 — Alex discussion)

- **Keep all roster rows** (no minutes filter).
- **Bench rule:** set **raw PPM = 0** for players with **under 20 minutes**, then z-score within season as usual.
- **Caps:** actual NCAA roster counts on the **full** panel.

Script flag: `--ppm-zero-below-minutes 20` in `pd21_rho_hsort_calibrate.py` (default min-20 filter unchanged; outputs use suffix `_ppm0lt20`).
What this simplifies: one roster universe for empirical vs simulated assignment — same bodies, same team sizes; bench carries no skill signal instead of being deleted. Fewer moving parts between "who is in emp H_{sort}" and "who is in sim LG."
What it does *not* simplify:

Layer	Still need empirical caps?	Still need a minutes policy?
ASSIGN ρ / H_{sort}	**Yes** — heterogeneous R_j	Test ppm-zero; may replace filter here
L_C / congestion diagnostics	**Yes**	Probably align with ASSIGN panel
Hero ventiles / LOO / draft MLE (λ, t)	Fixed empirical teams (no LG)	**Probably keep min $\geq$ 20** unless Alex says otherwise — draft estimand is about rotation contributors, not walk-ons

Score $\neq$ select (binding): ASSIGN can use full roster + zero bench while SELECT MLE uses a filtered hero panel. That is allowed and may be **correct** — different layers, different estimands.

If ppm-zero "works" (sensible interior ρ^*, not stuck at cap)

The story is likely **panel definition**, not "drop empirical caps":

> *Low homophily was partly because the minutes filter built a* ***different league*** *than raw NCAA rosters — sim at $\rho = 0$ was already "sorted enough" on the trimmed panel.*

That does **not** imply recalculating **everything** without filtering. Treat it **layer by layer** (see table above).

Question for Alex (next meeting)

> *"For draft MLE, do we keep `min_minutes = 20` on the hero panel even if ASSIGN ρ calibration uses full roster + zero bench?"*

Working guess: yes for MLE; maybe full roster + zero bench only for LG / H_{sort} — but Alex decides.

Option (a): Sorting-index calibration — match one statistic by simulation

Idea in plain language

Grandchild assignment (LG) is **random**. You cannot expect one simulation run to exactly reproduce one real season. Instead:
1. Measure the **real** sorting index (H_{sort}) from observed team labels.
2. For a trial value of ρ, run Grandchild assignment (LG) many times (50 seeds) with the **same** abilities and roster caps.
3. Average the simulated sorting index over those runs.
4. Pick the ρ where simulated and empirical sorting indices match.

You are **calibrating** the knob to a **readout**, not writing a full probability formula for the exact roster.

Tiny numeric example (made-up numbers)

Suppose **2015** has 4,000 players and empirical sorting index $H_{\text{sort}}^{\text{emp}} = 0.12$.

Trial ρ	50 simulation runs → mean $H_{\text{sort}}^{\text{sim}}$
0.3	≈ 0.07 (too mixed)
0.8	≈ 0.11 (close)
1.5	≈ 0.18 (too sorted)

Pick $\rho^* \approx 0.8$ for 2015.

Repeat for each season → ρ_t^* **per season**.
Then pick ρ_{long}^* by minimizing the same mismatch **averaged or summed across all seasons**.

What this is / is not

Is	Is not
Honest about Grandchild assignment (LG) being stochastic	A literal “probability this exact roster happened”
Easy to explain: “we match realized sorting”	Using draft data
Fast to implement (extends existing rho sweep script)	Matching global_wss or L_C unless you add them (→ option d)

Per-season vs longitudinal

- **Per season (ρ_t^*):** each year gets its own knob — “what homophily matches 2015 NCAA seating?”
- **Longitudinal (ρ_{long}^*):** one knob for 2011–2021 — “one ρ for the whole panel”

Both are natural. They can differ if sorting strength varied by era.

Option (b): Pseudo-likelihood on observed team labels

Idea in plain language

Grandchild assignment (LG) is **sequential** (players arrive one at a time; team centroids update). Real NCAA data gives only the **final** teams — **not** the seating order.

Pseudo-likelihood means: pretend each player’s team choice was a **simple independent decision** based on how close they are to each team’s talent, and ask: *“Under this simplified rule, how plausible are the teams we actually saw?”*

You **do not** re-run the simulator. You look directly at observed rosters.

The simplified rule (conceptual)

For player i, team j, with leave-one-out team mean $\bar{A}_{j,-i}$ (team j’s average ability **excluding** player i):

$$P(\text{player } i \text{ on team } j) \propto \exp\left(-\rho \cdot |\hat{A}_i - \bar{A}_{j,-i}|\right)$$

Multiply (or add log-) these probabilities for **every player’s actual team**. Pick ρ that makes the **observed** assignment score highest.

Tiny example: 6 players, 2 teams

Abilities: players 1–3 are high (10, 9, 8); players 4–6 are low (2, 3, 1).

Observed assignment: - Team A: players 1, 2, 4 → team mean ≈ 7.0 - Team B: players 3, 5, 6 → team mean ≈ 4.0

Question for player 4 (ability = 2, actually on Team A):

- Leave-one-out mean of Team A (without player 4): $(10 + 9)/2 = 9.5$
- Leave-one-out mean of Team B (without player 4): $(8 + 3 + 1)/3 = 4.0$

Player 4 is **closer** to Team B’s mean. Under **high** ρ, the simplified model says Team A is **surprising** for player 4.

- **Low ρ:** both teams look plausible → observed assignment not very surprising.
- **High ρ:** observed assignment looks **less** plausible unless ρ is tuned so distances work out.

Search for the ρ that makes the **whole season’s** observed labels as plausible as possible under this shortcut.

Why “pseudo”?

Because it **ignores**:

- Stub capacity dynamics (R_j shrinking as seats fill)
- Random seating order
- Endogenous centroids updating during the process

It is a **Besag-style** approximation (common in network statistics when the true model is too hard).

When it helps

Good for	Bad for
Fast sanity check: “is there a ρ consistent with labels?”	Being the **official** Grandchild assignment (LG) generative story
No seeds / no simulation loop	Matching L_C distribution (doesn’t simulate full rosters)
Cross-check against option (a)	Alex briefing if he asked for strict configuration-model MLE

Recommendation: run as **secondary audit**, not primary, unless Alex explicitly wants this as the main estimator.

Option (c): Exact sequential likelihood

Idea in plain language

Replay Grandchild assignment (LG) **exactly** as coded:

1. Shuffle player order (random permutation).

2. For each player in that order, compute weights $R_j \exp(-\rho|\hat{A}_i - \mu_j|)$ over open teams.
3. Normalize to probabilities, draw a team, update centroid μ_j.
4. Multiply the probability of **each step's choice** along that order.

For **empirical** data, you don't know which permutation NCAA "used." So you must either:

- **Average** over many random orders (expensive), or
- **Sum** over all permutations (impossible at NCAA scale).

Tiny example: 3 players, 2 teams (capacities 2 and 1)

Players: A=10, B=5, C=1. Team 1 has 2 seats, Team 2 has 1 seat.

One possible order: B, then C, then A.

- **Step 1 (B=5):** both teams open, centroids at league mean → B lands Team 1 with some probability (e.g. 0.55).
- **Step 2 (C=1):** Team 1 centroid updates → C lands somewhere with some probability.
- **Step 3 (A=10):** only open seats left; often forced or nearly forced.

Likelihood for **this order** = product of the three step probabilities.

Different order (A first, then B, then C) → **different** product, because centroids evolve differently.

For empirical fit: average likelihood over many orders, or treat order as random and integrate.

Tradeoffs

Pros	Cons
Closest to "true" Grandchild assignment (LG) generative model	**Slow** and fiddly
Principled if you handle order correctly	At NCAA scale (thousands of players), brutal without serious compute
	Still doesn't uniquely identify order for real data

Verdict for v1: skip unless Alex insists on full generative MLE. Options (a) and (d) capture the same ρ intent with far less pain.

Option (d): Multi-moment simulation matching — "stronger" criterion

Idea in plain language

Same engine as option (a) — run Grandchild assignment (LG) many times per ρ — but match **several** empirical summaries at once, not just sorting index (H_{sort}):

1. **Sorting index (H_{sort})** — how sorted are teams?
2. **Global within-team sum of squares (global_wss)** — raw spread around team means
3. **Team congestion (L_C) distribution** — histogram of team-level congestion (see `grandchild_empirical_lc_compare.py`)

Define a **combined loss**, e.g.:

$$\text{loss}(\rho) = w_1 \cdot |H_{\text{sort}}^{\text{sim}} - H_{\text{sort}}^{\text{emp}}| + w_2 \cdot |\text{global\_wss}^{\text{sim}} - \text{global\_wss}^{\text{emp}}| + w_3 \cdot \text{distance}(L_C^{\text{sim}}, L_C^{\text{emp}})$$

(simulated values = mean over 50 seeds)

Pick ρ that minimizes total loss — **per season** and **longitudinally**.

Tiny example (made-up)

Empirical 2015 targets:

- $H_{\text{sort}}^{\text{emp}} = 0.12$
- $\text{global\_wss}^{\text{emp}} = 8200$
- L_C distribution: peak near 0.45

At $\rho = 0.5$ (legacy default), 50 sim runs might give:

- $H_{\text{sort}}^{\text{sim}} = 0.09$ → error 0.03
- $\text{global\_wss}^{\text{sim}} = 9100$ → error 900
- L_C peak at 0.38 → histogram distance 0.05

At $\rho = 0.9$:

- $H_{\text{sort}}^{\text{sim}} = 0.12$ → error 0.00
- $\text{global\_wss}^{\text{sim}} = 8500$ → error 300
- L_C peak at 0.43 → distance 0.02

Combined loss might favor $\rho \approx 0.9$ even though $\rho = 0.5$ was "close enough" on sorting alone.

Why this matches the "stronger" success criterion

- Option (a) alone: "get sorting index right."
- Option (d): "get sorting **and** within-team spread **and** congestion shape right" — stronger claim that Grandchild assignment (LG) reproduces **structure**, not one number.

Recommended v1 primary, with option (b) as optional audit.

Side-by-side comparison

	(a) Sorting calibration	(b) Pseudo-likelihood	(c) Exact sequential LL	(d) Multi-moment sim match
Uses simulator?	Yes, many seeds	No	Yes, many orders	Yes, many seeds
Uses observed team labels directly?	Only for target stats	Yes, in formula	Yes, in path	Only for target stats
Handles randomness?	Average over seeds	N/A	Average over orders	Average over seeds
Matches H_{sort}?	Yes	Indirectly	Yes	Yes
Matches global_wss / L_C?	Not unless added	No	Can in principle	**Yes**
Implementation effort	Low	Low	High	Medium
Alex story	"Calibrate ρ to sorting"	"Approximate label likelihood"	"Full generative MLE"	"Calibrate ρ to roster geometry"

How this connects to implementation choices

Choice	Meaning
50 seeds	For (a) and (d), each (ρ, season) runs Grandchild assignment (LG) 50 times; report mean ± spread
One ρ per season + longitudinal ρ	Grid search inside each season; then one pooled objective across 2011–2021
$\rho = 0.5$ reference	Always report the same three metrics at legacy default
HPC / Rivanna	Not required to **build** the script; parallelize $(\mathrm{season}, \rho, \mathrm{seed})$ when seeds or grid fineness increase

Planned outputs (when coded): `HEROs_and_PASSes/pd21_rho/` — script, JSON, plots.

Suggested path forward

Role	Option
Primary	**(d)** — multi-moment simulation matching
Audit	**(b)** — pseudo-likelihood, reported alongside

One-liner for Alex:

> *"We calibrate Grandchild assignment's ρ so simulated rosters match empirical sorting, within-team spread, and L_C shape — per season and pooled — not from draft likelihood. Draft MLE for λ and t stays on fixed empirical teams."*

Charles runs PDF when ready:

```
./scripts/convert_single_md_to_pdf.sh 3-Master_Plan/Alex_notes/rho_est_options_for_dummies.md
```